

# CS3.301 - Assignment 1

Emmanuel Roshan Jaiby, 2025101010

## Q1.

(a) Let  $\Sigma = \{a, b\}$ . For each  $k \geq 1$ , let  $C_k$  be the language consisting of all strings that contain an  $a$  exactly  $k$  places from the right-hand end. Describe an NFA with  $k + 1$  states that recognizes  $C_k$ , both in terms of a state diagram and a formal description.

(b) Consider the languages  $C_k$  defined in the previous problem. Prove that for each  $k$ , no DFA can recognize  $C_k$  with fewer than  $2^k$  states.

**Sol.** Let  $M_k$  be the NFA recognising  $C_k$ . Then  $M_k$  is given by the 5-tuple  $(Q, \Sigma, \delta, q_0, F)$ , where:

$$\begin{aligned} Q &= \{q_0, \dots, q_k\}, \\ \Sigma &= \{a, b\}, \\ q_0 &\in Q \text{ is the start state,} \\ F &= \{q_k\}, \end{aligned}$$

and  $\delta : Q \times \Sigma \rightarrow \mathcal{P}(Q)$  is given by:

$$\begin{aligned} \delta(q_0, a) &= \{q_0, q_1\} \\ \delta(q_0, b) &= \{q_0\} \\ \delta(q_i, a) &= \{q_{i+1}\} \text{ for all } 1 \leq i \leq k-1 \text{ and } k \geq 2 \\ \delta(q_i, b) &= \{q_{i+1}\} \text{ for all } 1 \leq i \leq k-1 \text{ and } k \geq 2 \end{aligned}$$

This NFA loops at  $q_0$  until it guesses non-deterministically when it reads an  $a$  that it is  $k$  places from the right-hand end. It then transitions to  $q_1$ . All further states simply serve to count the  $k$  places from the right-hand end.

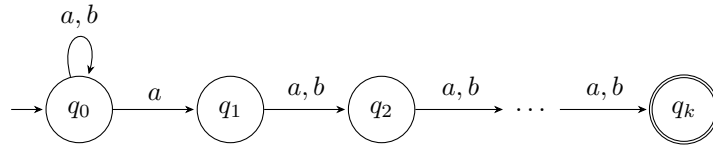


Figure 1: NFA  $M_k$  recognizing  $C_k$ .

Intuitively, a DFA that recognizes  $C_k$  must be able to "remember" the last  $k$  symbols. Since there are  $2^k$  such strings, if the DFA has fewer than  $2^k$  states, then there would be some state  $q$  that can be reached by two different strings of length  $k$ .

Let these two strings be  $x$  and  $y$ .

**Case 1.** If they differ at the first position, then one of them must start with a 0 and one must start with a 1. By symmetry, let  $x$  start with a 0 and  $y$  start with a 1.

Then,  $x$  is a string that does not have a 1  $k$  places from the right-hand end, and hence  $q \notin F$ . However,  $y$  is a string that has a 1  $k$  places from the right-hand end and hence  $q \in F$ . This is a contradiction.

**Case 2.** If they differ at some position  $i \neq 1$ , then suffix both  $x$  and  $y$  with  $i - 1$  zeroes. Now, one can take the suffixes  $x'$  and  $y'$  of length  $k$  of  $x$  and  $y$  respectively.  $x'$  and  $y'$  differ at the first position and lead to the same state  $q$ . Now, by the argument in the previous case, there is a contradiction.

Hence, a DFA recognizing  $C_k$  must have  $2^k$  states. ■

## Q2.

The Hamming distance between two words of equal length is the number of positions at which they differ. Prove that for every regular language  $L$  and every constant  $k$ , the set of words at a distance at most  $k$  from a word in  $L$  is regular.

**Sol.** Since  $L$  is regular, there is a DFA  $M$  that recognizes it, given by  $(Q, \Sigma, \delta, q_0, F)$ . We construct an NFA  $M_k$  that recognises the language of all strings that are at a distance of at most  $k$  from any word in  $L$  as

$$(Q \times \{0, 1, \dots, k\}, \Sigma, \delta_k, (q_0, 0), F \times \{0, 1, \dots, k\})$$

and  $\delta_k : Q \times \{0, 1, \dots, k\} \times \Sigma \rightarrow \mathcal{P}(Q \times \{0, 1, \dots, k\})$  is given by:

$$\delta_k((q, i), a) = \{(\delta(q, a), i)\} \cup \{(\delta(q, b), i + 1) : b \in \Sigma \setminus \{a\}\} \quad i < k$$

$$\delta_k((q, k), a) = \{(\delta(q, a), k)\} \cup \emptyset$$

$M_k$  non-deterministically guesses whether the character it reads is correct or wrong. The correct case corresponds to  $\{(\delta(q, a), i)\}$  and the wrong case corresponds to the set of states with the error counter  $i$  incremented and all other characters considered to be possibly right. When  $k$  errors have accumulated and  $M_k$  guesses the character read is "wrong", it enters a dead state  $\emptyset$ .

If the input string has no errors, since it would have reached a state  $q \in F$  in the original DFA  $M$ , it will reach  $(q, 0)$  on at least one run, and hence be accepted.

If the input string has no more than  $k$  errors, on at least one run,  $M_k$  at every error will guess the "correct" character in its place, and reach an accepting state  $(q, n)$ , where  $n \leq k$ .

If the input string has more than  $k$  errors, then  $M_k$ , after recognising and correcting the first  $k$  errors on all runs, must send it to the dead state on the  $(k + 1)^{\text{th}}$  error or assume it is correct and then fail or crash.

■

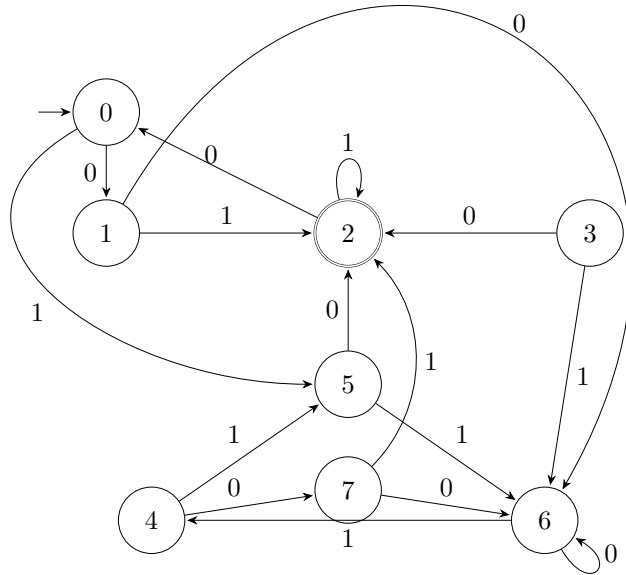


Figure 2: DFA

### Q3.

Minimize the DFA given below.

**Sol.** We first eliminate all states unreachable from the start state, i.e., state 3. We then separate the remaining states into two groups —accepting and non-accepting states.

$$\{0, 1, 4, 5, 6, 7\} \{2\}$$

These states, upon reading 0, go to the following respectively:

$$\{1, 6, 7, 2, 6, 6\} \{1\}$$

Since state 5 on a read of 0 is accepting and the states 0, 1, 4, 6, 7 are not, state 5 is distinguished from them.

$$\{0, 1, 4, 6, 7\} \{2\} \{5\}$$

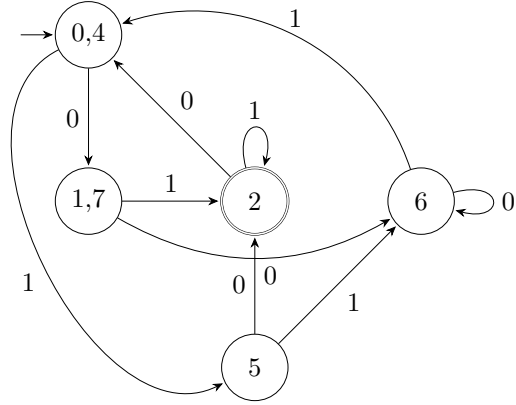


Figure 3: Minimised DFA

These states, upon reading 1, go to the following respectively;

$$\{5, 2, 5, 4, 2\} \{2\} \{6\}$$

Depending on which group they go to, the pair of states (0,4) and (1,7) are distinguished from each other.

$$\{0, 4\} \{1, 7\} \{2\} \{5\} \{6\}$$

The pair (1,7) are indistinguishable since they go to states 2 and 6 upon reading 1 and 0, respectively. The pair (0,4) are indistinguishable since, upon reading 0, they go to (1,7) respectively, which is an indistinguishable pair, and upon reading 1, they both go to state 5. Hence, the minimal DFA is given by the following:

#### Q4.

Let  $\Sigma = \{a\}$  be a single-letter alphabet. Prove that a language  $L \subseteq \Sigma^*$  is regular if and only if the set of the lengths of its words,  $S = \{|w| \mid w \in L\}$ , is ultimately periodic.

**Def.** A set of natural numbers  $S \subseteq \mathbb{N}$  is ultimately periodic if there exists a threshold integer  $N \geq 0$  and a period integer  $p \geq 1$  such that for all integers  $n \geq N$ :

$$n \in S \iff n + p \in S$$

**Sol.**

**Forward Direction.** Let the lengths of words of  $L$  be ultimately periodic. Then, there exists a threshold integer  $N \geq 0$  and a period integer  $p \geq 1$ . Then,  $L$  is a union of regular languages given by:

$$L = \{a^n \mid n \in S, n < N\} \cup \bigcup_{i \in I} \{a^{N+i+kp} \mid k \geq 0\}$$

where  $I = \{i \in \{0, 1, \dots, p-1\} \mid N+i \in S\}$  and is hence itself finite under closure of finite union, since the first term in the union is a finite language and the second is represented as a union of regular languages  $a^{N+i}(a^p)^*$ .

**Backward Direction.** Let  $D$  be the DFA that recognizes  $L$ . Since  $D$  is a finite automaton, it must have a finite number of states.

If  $L$  is finite, then there is a maximum length  $\max(S)$ . Let the threshold integer  $N = \max(S) + 1$  and the period integer  $p = 1$ . Then,  $\forall n \geq N, n \in S \iff n + p \in S$  is vacuously true. Hence, the set of the lengths of the words of  $L$  is ultimately periodic.

If  $L$  is infinite, then by the pigeonhole principle, there must be strings that end at the same state despite being of different lengths, since for any number of states of the DFA recognizing  $L$ , there must be strings of length greater than that number, since otherwise  $L$  would be finite. Consider the sequence of states  $q_0, q_1, \dots$  where  $q_k = \hat{\delta}(q_0, a^k)$ . By the pigeonhole principle, there are indices  $m, m+p$  such that  $q_m = q_{m+p}$ . For integers  $n \geq m$ ,  $a^{n+p} = a^{m+p}a^{n-m} = a^m a^{n-m} = a^n$ . Since  $\hat{\delta}(q_0, a^m) = q_m$  and  $\hat{\delta}(q_m, a^p) = q_{m+p} = q_m$ ,  $\hat{\delta}(q_0, a^{n+p}) = \hat{\delta}(q_0, a^n) = q_n$ . Hence,  $a^n \in L \iff q_n \in F \iff q_{n+p} \in F \iff a^{n+p} \in L$ . Equivalently,  $n \in S \iff n + p \in S$ . ■

## Q5.

Consider the language  $L = \{a^i b^j | i \neq j\}$ .

- (a) Prove that  $L$  is not regular using the pumping lemma.
- (b) Provide a much shorter proof that  $L$  is not regular using only closure properties and the known non-regular language  $\{a^n b^n | n \geq 0\}$ .

**Sol.**

**a.** Assume to the contrary that  $L$  is regular. Let  $p$  be the pumping length given by the pumping lemma. Consider a string  $w = a^p b^{p+p!} \in L$ . Now, by the pumping lemma, we can break  $w$  into three strings  $xyz$  such that  $y \neq \epsilon$ ,  $|xy| \leq p$ , and  $\forall k \geq 0, xy^k z \in L$ . The largest possible  $xy$  is  $a^p$ . Choose  $y = a^m$ ,  $1 \leq m \leq p$ . Now,  $\forall k \geq 0, xy^k z \in L$ . Choose  $k = 1 + \frac{p!}{m}$ . Then,

$$xy^k z = a^{p+p!} b^{p+p!}$$

, which is not in  $L$ . Hence,  $L$  is non-regular. ■

**b.** Assume to the contrary that  $L$  is regular. Let  $R$  be the regular language described by the regular expression  $a^* b^*$ .  $R$  is regular since both  $a^*$  and  $b^*$  are regular languages by closure under the star operation and closure under the concatenation operation. Since  $L$  is regular, by closure under complement,  $\bar{L}$  is regular.  $R \setminus L$  is regular since  $R \setminus L = R \cap \bar{L} = \overline{\bar{R} \cup L}$  is closed under complementation and union.  $R \setminus \bar{L}$  consists of those strings of zero or more  $a$ 's followed by zero or more  $b$ 's such that the number of  $a$ 's is not unequal to the number of  $b$ 's. Hence,  $R \setminus L = \{0^n 1^n | n \geq 0\}$ . But this is a contradiction since  $\{0^n 1^n | n \geq 0\}$  is a non-regular language. Hence  $L$  must be non-regular. ■

## Q6.

Let

$$\Sigma_3 = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

$\Sigma_3$  contains all size-3 columns of 0s and 1s. A string of symbols in  $\Sigma_3$  gives three rows of 0s and 1s. Consider each row to be a binary number written with the least significant bit first (i.e. the leftmost column is the units bit) and let

$$B = \{w \in \Sigma_3^* \mid \text{the bottom row of } w \text{ is the sum of the top two rows}\}.$$

For example,

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \in B, \quad \text{but} \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \notin B.$$

Show that  $B$  is regular.

**Sol.** We construct a DFA  $D_B = (Q, \Sigma_3, \delta, q_0, F)$  to recognise the language  $B$  by:

$$\begin{aligned} Q &= \{q_0, q_1, q_X\}, \\ q_0 &\in Q \text{ is the start state,} \\ F &= \{q_0\}, \end{aligned}$$

and  $\delta : Q \times \Sigma_3 \rightarrow Q$  is given by the transition diagram. The dead state  $q_X$  is transitioned to when the input string, no matter the suffix, cannot possibly be in  $B$ .  $q_0$  represents a valid string in  $B$  with carry 0, and  $q_1$  represents a valid string satisfying the property that the last row is the sum of the first two that has a carry of 1. If, in  $q_0$ , despite having no carry, a symbol is read that would require carry 1 to be valid, or vice-versa in  $q_1$ , it is to  $q_X$  that a transition occurs.

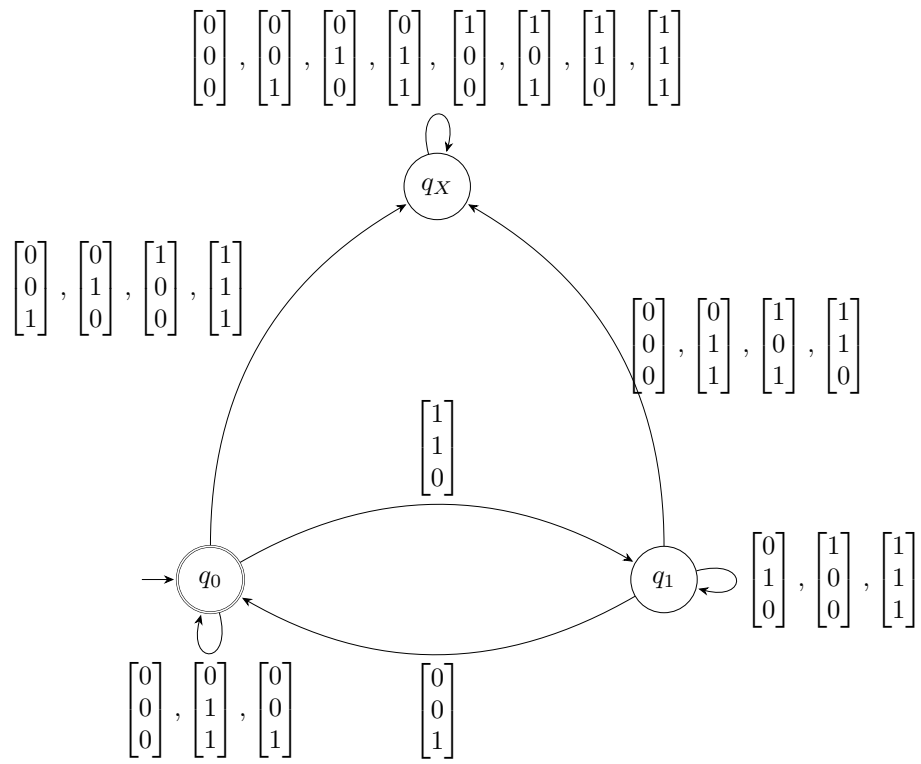


Figure 4: Transition diagram for  $\delta$

## Q7.

Prove that the following languages are not regular:

- (a)  $\{a^{2^n} : n \in \mathbb{N}\}$
- (b)  $\{a^p : p \text{ is a prime number.}\}$
- (c)  $\{a^i b^j : \gcd(i, j) = 1\}$
- (d)  $\{\text{bin}(p) : p \text{ is a prime number.}\}$

**Sol.**

**a.** Assume, to the contrary, that  $L = \{a^{2^n} : n \in \mathbb{N}\}$ . Let  $p$  be the pumping length specified by the pumping lemma. Consider the string  $a^{2^p} \in L$ . Since  $a^{2^p}$  is composed solely of  $a$ 's,  $xy = a^p$ . Let  $y = a^m$ , where  $1 \leq m \leq p$ . By the pumping lemma, all strings of the form  $xy^i z \in L$ . Let  $i = 2$ . Then,  $xy^2 z = a^{2^p + m}$ .  $2^p < 2^p + 1 \leq 2^p + m \leq 2^p + p < 2^{p+1}$ , since  $2^x > x \forall x \in \mathbb{R}$ . Hence,  $2^p + m$  is between two consecutive powers of two and cannot be a power of two. This is a contradiction, Hence,  $L$  is non-regular. ■

**b.** Assume, to the contrary, that  $L = \{a^p : p \text{ is a prime number.}\}$  is a regular language. Let  $q$  be the pumping length specified by the pumping lemma. Consider the string  $a^n$ , where  $n$  is the smallest prime strictly greater than  $q$ . Now, the pumping lemma states that  $a^n$  can be split into three strings  $xyz$  such that

$$xyz = a^n, |y| > 0, |xy| \leq q, xy^i z \in L \quad \forall i \geq 0$$

. Let  $|y| = m > 0$  and  $i = n + 1$ . Then,  $xy^i z = a^{n+nm}$ .  $n + nm = n(1 + m)$ , which is composite. Hence,  $xy^i z \notin L$ . This is a contradiction. Hence,  $L$  is non-regular. ■

**c.** Assume, to the contrary, that  $L = \{a^i b^j : \gcd(i, j) = 1\}$  is a regular language. Let  $p$  be the pumping length specified by the pumping lemma. Consider the string  $a^{q-1} b^q$ , where  $q$  is a prime strictly greater than  $2p$ , which is in  $L$  since  $p$  has only 1 and  $q$  as factors, and since  $(-1)(q-1) + (1)q = 1$ , by Bézout's lemma,  $\gcd(q-1, q) = 1$ . Now, the pumping lemma states that  $a^n$  can be split into  $C$ . Let  $|y| = m$ . Since  $q$  is prime, by Bézout's lemma,  $\exists s, t \in \mathbb{Z}$  s.t.  $sm + tq = \gcd(m, q) = 1$ , since  $m \leq p < q$ . Let  $r$  be the smallest integer such that  $s + rq > 0$ . Let  $i = 1 + s + rq$ . Then,  $xy^i z = a^{p-m} (a^m)^{1+s+rq} a^{q-p-1} b^q = a^{q-1+m(s+rq)} b^q$ . Now,  $q-1+m(s+rq) = -1+1+0 \pmod{q}$ , since  $ms = 1 - tq \equiv 1 \pmod{q}$ . So  $q \mid q-1+m(s+rq)$  and hence  $\gcd(q, q-1+m(s+rq)) \neq 1$ . This is a contradiction. Hence  $L$  is non-regular. ■

**d.** Assume, to the contrary, that  $L = \{\text{bin}(p) : p \text{ is a prime number.}\}$  is regular. Then, there is a DFA  $D$  that recognizes  $L$ . Then, let  $k$  be the number of states in  $D$ . Then,  $k$  is also the pumping length specified by the pumping

lemma. Let  $p$  be a prime whose binary representation has more than  $k$  digits. Let  $\text{bin}(p)$  be split into three strings  $xyz$  per the pumping lemma. Let  $p_i$  be the value of the pumped string  $xy^iz$ .

$$p_i = \text{val}(x)2^{im+n} + \text{val}(y)2^n(1 + 2^m + 2^{2m} + \dots + 2^{(i-1)m}) + \text{val}(z)$$

Hence,

$$p_i = \text{val}(x)2^{im+n} + \text{val}(y)2^n \frac{2^{im} - 1}{2^m - 1} + \text{val}(z)$$

. Since  $p = p_1$  is odd (since if  $D$  had only one state, only then would  $p$  possibly equal two, but then  $D$  must accept all inputs or reject all inputs, and it would not then recognize  $L$ , hence it must have at least two states, i.e.,  $p$  is at least 5),  $\gcd 2, j = 1$ . Hence, by Fermat's little theorem,  $2^{p-1} \equiv 1 \pmod{p}$ . Consider

$$p_p = \text{val}(x)2^{pm+n} + \text{val}(y)2^n \frac{2^{pm} - 1}{2^m - 1} + \text{val}(z) \equiv \text{val}(x)2^{m+n} + \text{val}(y)2^n + \text{val}(z) \equiv p \equiv 0 \pmod{p}$$

, since  $2^{pn} = 2^{n+(p-1)n} = 2^n(2^n)^{p-1} \equiv 2^n(1) \pmod{p}$ . Hence, the pumped string is composite. This contradicts the assumption that  $L$  is regular. Hence,  $L$  is non-regular. ■.

## Q8.

Prove that the following grammars are ambiguous by providing a string with two different leftmost/rightmost derivations, or two different parse trees:

(a)

$$S \rightarrow aAB$$

$$A \rightarrow bBb$$

$$B \rightarrow A|\lambda$$

(b)

$$S \rightarrow SS|\lambda|aSb|bSa$$

(c)

$$S \rightarrow aSb|SS|\lambda$$

**Sol.**

**a.** Consider the string *abbbb*.

**Derivation 1.**  $S \rightarrow aAB$

$aAB \rightarrow abBbA$  (by  $A \rightarrow bBb$ )

$abBbA \rightarrow abbA$  (by  $B \rightarrow \lambda$ )

$abbA \rightarrow abbbBb$  (by  $A \rightarrow bBb$ )

$abbbBb \rightarrow abbbb$  (by  $B \rightarrow \lambda$ )

**Derivation 2.**  $S \rightarrow aAB$

$aAB \rightarrow abBb$  (by  $A \rightarrow bBb$ )

$abBb \rightarrow abAb$  (by  $B \rightarrow A$ )

$abAb \rightarrow abbBbb$  (by  $A \rightarrow bBb$ )

$abbBbb \rightarrow abbbb$  (by  $B \rightarrow \lambda$ ).

Since there are 2 leftmost derivations of the string *abbbb*, the CFG is ambiguous.

■

**b.** Consider the string *ab*.

**Derivation 1.**  $S \rightarrow SS$

$SS \rightarrow aSbS$  (by  $S \rightarrow aSb$ )

$aSbS \rightarrow abS$  (by  $S \rightarrow \lambda$ )

$abS \rightarrow ab$  (by  $S \rightarrow \lambda$ ).

**Derivation 2.**  $S \rightarrow aSb$

$aSb \rightarrow ab$  (by  $S \rightarrow \lambda$ ).

Since there are 2 leftmost derivations of the string *ab*, the CFG is ambiguous.

■

c. Consider the string  $ab$ .

**Derivation 1.**  $S \rightarrow SS$

$SS \rightarrow aSbS$  (by  $S \rightarrow aSb$ )

$aSbS \rightarrow abS$  (by  $S \rightarrow \lambda$ )

$abS \rightarrow ab$  (by  $S \rightarrow \lambda$ ).

**Derivation 2.**  $S \rightarrow aSb$

$aSb \rightarrow ab$  (by  $S \rightarrow \lambda$ ).

Since there are 2 leftmost derivations of the string  $ab$ , the CFG is ambiguous.

■

## Q9.

Construct a PDA for each of the following languages.

(a)

$$L = \{a^m b^n \mid m \leq n \leq 2m\}$$

(b)

$$L = \{a^i b^j c^k \mid i = j \text{ or } j = k\}$$

(c) For any one of the above parts, if you constructed a PDA which accepts by final state, convert it into a PDA which accepts by empty stack instead; and vice-versa if you did it the other way around.

**Sol.**

a. Let  $M$  be the PDA given by  $(Q, \Sigma, \Gamma, \delta, q_0, F)$  where:

$$\begin{aligned} Q &= \{q_a, q_1, q_b, q_f\} \\ \Sigma &= \{a, b\} \\ \Gamma &= \{X\} \\ F &= \{q_f\} \end{aligned}$$

and  $\delta : Q \times \Sigma_\epsilon \times \Gamma_\epsilon \rightarrow \mathcal{P}(Q \times \Gamma_\epsilon)$  is given by:

$$\begin{aligned} \delta(q_0, a, \epsilon) &\rightarrow \{(q_0, a), (q_1, a)\} \\ \delta(q_1, \epsilon, \epsilon) &\rightarrow \{(q_0, a)\} \\ \delta(q_0, \epsilon, \$) &\rightarrow \{(q_f, \$)\} \\ \delta(q_0, b, a) &\rightarrow \{(q_1, \epsilon)\} \\ \delta(q_1, b, a) &\rightarrow \{(q_1, \epsilon)\} \\ \delta(q_1, \epsilon, \$) &\rightarrow \{(q_f, \$)\} \end{aligned}$$

and we assume  $\$$  is on the stack before execution starts. While reading  $a$ 's,  $M$  pushes one or two  $a$ 's onto the stack, non-deterministically. When it reads  $b$ 's, it pops one off the stack. If the string is empty, i.e.,  $m = n = 0$ , or when the end of the input string is reached, if  $M$  is in the final state  $q_f$ , it accepts, if the stack marker  $\$$  is on top to signal it is effectively empty.

b. Let  $M$  be the PDA given by  $(Q, \Sigma, \Gamma, \delta, q_0, F)$  where:

$$\begin{aligned} Q &= \{q_0, q_{1a}, q_{1b}, q_{1c}, q_{2a}, q_{2b}, q_{2c}, q_f\} \\ \Sigma &= \{a, b, c\} \\ \Gamma &= \{X\} \\ F &= \{q_f\} \end{aligned}$$

and  $\delta : Q \times \Sigma_\epsilon \times \Gamma_\epsilon \rightarrow \mathcal{P}(Q \times \Gamma_\epsilon)$  is given by:

$$\delta(q_0, \epsilon, \$) \rightarrow \{(q_{1a}, \$), (q_{2a}, \$)\}$$

$$\begin{aligned}
\delta(q_{1a}, a, \epsilon) &\rightarrow \{(q_{1a}, X)\} \\
\delta(q_{1a}, b, X) &\rightarrow \{(q_{1b}, \epsilon)\} \\
\delta(q_{1b}, b, X) &\rightarrow \{(q_{1b}, \epsilon)\} \\
\delta(q_{1c}, c, \$) &\rightarrow \{(q_{1c}, \$)\} \\
\delta(q_{1c}, \epsilon, \$) &\rightarrow \{(q_f, \$)\} \\
\delta(q_{2a}, a, \$) &\rightarrow \{(q_{2a}, \$)\} \\
\delta(q_{2a}, b, \$) &\rightarrow \{(q_{2b}, X)\} \\
\delta(q_{2b}, b, \epsilon) &\rightarrow \{(q_{2b}, X)\} \\
\delta(q_{2c}, c, X) &\rightarrow \{(q_{2c}, \epsilon)\} \\
\delta(q_{2c}, \epsilon, \$) &\rightarrow \{(q_f, \$)\}
\end{aligned}$$

and we assume \$ is on the stack before execution starts. Initially, the PDA decides whether to test for  $i = j$  or  $j = k$ . These represent the set of states whose subscript includes a 1 or a 2 respectively. In the first branch,  $X$  is pushed when  $a$  is read and popped when  $b$  is read, with no interaction with the stack on reading  $c$ . In the first branch,  $X$  is pushed when  $b$  is read and popped when  $c$  is read, with no interaction with the stack on reading  $a$ . The transition to the final state is only done when the stack marker \$ is on top, signalling the stack is effectively empty, and then the acceptance in final state triggers only if the input string's end has been reached.

**c.** We convert the PDA  $M = (Q, \Sigma, \Gamma, \delta, q_0, F)$  defined in part (a) into an empty-stack-accepting PDA  $M_e = (Q \cup \{q_s, q_e\}, \Sigma, \Gamma \cup \{\$, \delta_e, q_s, \emptyset\})$  as follows:

$$\begin{aligned}
\delta_e(q_s, \epsilon, \$_0) &= \{(q_0, \$\$_0)\} \\
\delta_e &= \delta \text{ for all inputs accepted by } \delta \\
\delta_e(q, \epsilon, \gamma) &= \{q_e, \epsilon\}, \forall q \in F, \gamma \in \Gamma \\
\forall \gamma \in \Gamma \cup \{\$, \delta_e(q_e, \lambda, \gamma) &= \{(q_e, \epsilon)\}
\end{aligned}$$

and we assume \$ is on the stack before execution starts. We add a new stack symbol, and allow the PDA to accept by final state by redirecting to an erase state  $q_e$  that pops all elements on the stack. The emptying of all elements on the stack allows the PDA to accept.

### Q10.

If  $A$  is any language, let  $A_{1/3-1/3}$  be the set of all strings in  $A$  with their middle thirds removed such that  $A_{1/3-1/3} = \{xz \mid \text{for some } y, |x| = |y| = |z|, xyz \in A\}$ . Show that, if  $A$  is regular, then  $A_{1/3-1/3}$  is not necessarily regular.

**Sol.** We construct the language  $A$  as the regular expression  $a^*bc^*$  over the alphabet  $\Sigma = \{a, b, c\}$ .  $A$  is described by a regular expression and is hence a regular language.

Let  $L = A_{1/3-1/3}$ . Assume, to the contrary, that  $L$  is regular. Since regular languages are closed under intersection,  $L' = L \cap a^*c^*$  is regular.

Let  $x = a^n, y = bc^{n-1}, z = c^n$ . Since the lengths are identical and  $xyz = a^nb c^{2n-1}$ , this string is in  $A$ . Hence,  $xz = a^nc^n \in L$ . Since it is also of the form  $a^*b^*$ , it is in  $L'$ .

Let  $w \in L'$ . By definition,  $w = xz$  of equal length  $x, z$  and another string of the same length  $y$  makes  $xyz \in A$ . Now,  $w$  contains no  $b$ , so the  $b$  by elimination is in  $y$ . This means  $xyz$  has length at least 3. Because  $x$  appears before  $y$ ,  $x = a^n$ . Similarly,  $z = c^n$ . Hence,  $w = \{a^nc^n \mid n > 0\}$ .

Now,  $L' = \{a^nc^n \mid n > 0\}$ , which is non-regular. But if this were the case, then  $A_{1/3-1/3}$  must have been non-regular, since otherwise closure under intersection would not hold. ■.